MAT237 Summary

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Part I

Foundations

In this section, we define notation and characterize sets and functions. We denote vectors as $\mathbf{v} \in \mathbb{R}^n$ and scalars as $c \in \mathbb{R}$. Note that vectors and scalars are treated distinctly, even in the case where $n = 1$.

§1 Curves and Functions

A more detailed look at parametric curves comes later in the vector calculus section. For now, a curve is some function $\gamma : I \to \mathbb{R}^n$, with $I \subseteq \mathbb{R}^n$.

Definition 1.1. Some terminology: at time $t \in I$, define

Position	$\gamma(t)$
Velocity	$\gamma'(t) := \lim_{h \to 0} \dfrac{\gamma(t+h) - \gamma(t)}{h}$
Acceleration	$\gamma''(t) := \lim_{h \to 0} \dfrac{\gamma'(t+h) - \gamma'(t)}{h}$
Speed	$\lVert\gamma'(t)\rVert$
Unit Tangent	$T(t) = \frac{\gamma'(t)}{\lVert\gamma'(t)\rVert}$
Unit Normal	$N(t) = \frac{T'(t)}{\lVert T'(t)\rVert}$
Unit Binormal	$B(t) = T(t) \times N(t)$
Trace	$\gamma(I)$

We can write γ in terms of its components. $\gamma = (\gamma_1, \ldots, \gamma_n)$ with each $\gamma_i : \mathbb{R} \to \mathbb{R}$. When differentiating, we can differentiate component-wise, since $\gamma'(t) = (\gamma_1'(t), \ldots, \gamma_n'(t))$.

When both $T(t)$ and $N(t)$ exist, they are orthogonal, as in $T(t) \cdot N(t) = \mathbf{0}$. Indeed,

$$\begin{aligned} \lVert T(t)\rVert &= 1 \\ \frac{d}{dt}[T(t) \cdot T(t)] &= \frac{d}{dt}[1] \\ 2T(t) \cdot T'(t) &= 0 \end{aligned}$$

Since $T(t)$ and $N(t)$ are orthogonal, get a unique **unit binormal** through $B(t) = T(t) \times N(t)$.

We now work with functions $f : A \to \mathbb{R}$, $A \subseteq \mathbb{R}^n$.

Definition 1.2 (k-Level Set)**.** For $k \in \mathbb{R}$, the level set of f at k is $\{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) = k\}$

Definition 1.3 (Graph)**.** $\text{graph}(f) = \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in A\}$

Definition 1.4 (Slices)**.** The i-coordinate slice of the graph of f at $c \in \mathbb{R}$ is $\text{graph}(f\big|_S)$, with $S = \{\mathbf{x} \in A : x_i = c\}$.

The ideas here generalize to functions $f : A \to \mathbb{R}^m$. The analogue to k-level sets is $f^{-1}(\{\mathbf{k}\})$.

Coordinate transformations and vector fields both relate to functions $f : \mathbb{R}^n \to \mathbb{R}^n$, covered in the differential and vector calculus portion respectively.

§2 Topology

Let $A \subseteq \mathbb{R}^n$, $\varepsilon \in \mathbb{R}$, $\varepsilon > 0$. For some point $\mathbf{p} \in \mathbb{R}^N$, define the ε-ball at $\mathbf{p}$ as

$$B_\varepsilon(\mathbf{p}) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{p}\| < \varepsilon\}$$

Say that ε the radius of the ball.

We want to characterize some point $\mathbf{p} \in \mathbb{R}^n$ relative to $A \subseteq \mathbb{R}^n$.

Definition 2.1.

Interior Point $\mathbf{p}$ is an interior point of A if there exists $\varepsilon \in \mathbb{R}$, $\varepsilon > 0$ with $B_\varepsilon(\mathbf{p}) \subseteq A$.

Boundary Point $\mathbf{p}$ is a boundary point of A if for every $\varepsilon \in \mathbb{R}$, $\varepsilon > 0$ there exists points $\mathbf{a}, \mathbf{b} \in B_\varepsilon(\mathbf{p})$ with $\mathbf{a} \in A$, $\mathbf{b} \notin A$.

Limit Point $\mathbf{p}$ is a limit point of A if for every $\varepsilon \in \mathbb{R}$, $\varepsilon > 0$, exists a point $\mathbf{q} \in B_\varepsilon(\mathbf{p}) \cap A$ with $\mathbf{p} \neq \mathbf{q}$.

The set of all interior points, all boundary points, and all limit points of A are respectively written A^o, ∂A, A^*.

The closure of A, written $\overline{A}$, is $A \cup A^*$.

Theorem 2.2: Let $A, B \subseteq \mathbb{R}^n$.
Have $A \subseteq \overline{A}$, $\overline{A \cup B} = \overline{A} \cup \overline{B}$, $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$, $\overline{A \times B} = \overline{A} \times \overline{B}$.
Furthermore, $A^o \subseteq A \subseteq \overline{A}$, $A^o \cap \partial A = \emptyset$, $\overline{A} = A^o \cup \partial A, \partial A = \overline{A} \setminus A^o$.